PG 1: FOUNDATIONS & UNIVARIATE TESTS (COURS 2, 3)

KEY DEFINITIONS: MULTIVARIATE & ML

Eigenvalue: Amount of variance explained by a specific Principal Component.

Orthogonal: Perpendicular / Uncorrelated (Angle = 90 deg).

Inertia: Term for Variance in Factorial Analysis.

Centroid: The geometric center of a cluster (Mean position of points).

Supervised Learning: Data has labels/answers (Y is known). e.g., Regression, LDA.

Unsupervised Learning: No labels. Finding structure. e.g., PCA, K-Means.

Overfitting: Model learns "noise" instead of signal. High accuracy on Train, low on Test.

Pruning: Cutting back a Decision Tree to prevent overfitting.

KEY DEFINITIONS: MODELING

Residuals: Difference between Actual value (Y) and Predicted value (Y-hat).

R-Squared: % of variance in Y explained by X. (Higher is usually better).

Homoscedasticity: Constant variance of residuals (Required for Linear Models).

Multicollinearity: When predictors (Xs) are correlated with each other (Bad).

Interaction: When the effect of Factor A changes depending on Factor B.

Maximum Likelihood (MLE): Algorithm used in GLM to find parameters that maximize prob of data.

Link Function: Function connecting the linear predictor to the mean of the distribution (e.g., Logit).

Overdispersion: When Variance > Mean in Poisson data (Need Negative Binomial).

1. Hypothesis Testing Core

H0 (Null): No effect / No difference / Randomness.

H1 (Alt): Effect exists / Difference exists.

p-value: Prob. of observing results if H0 is true.

Rule: If p < α (0.05) → **Reject H0** (Significant).

Rule: If p > α (0.05) → **Fail to reject H0** (Not sig).

Alpha (α): Type I error (False Positive). usually 5%.

Beta (β): Type II error (False Negative).

Power (1-β): Ability to detect an effect if it exists. Increases with N.

2. Descriptive Stats & R Basics

Measures: Mean (Sensitive to outliers), Median (Robust), SD (Spread).

Skewness: 0=Normal, >0 Right tail (Mean>Med), <0 Left tail.

Kurtosis: Tails weight. Normal=3 (or 0 excess).

summary(df): Quick stats for all cols.

table(df$cat): Counts for factors.

prop.table(): Convert counts to %.

3. Crucial Assumptions (Check FIRST)

A. Normality (Gaussian)

Visual: Histogram (Bell curve), Q-Q Plot (Points on line).

Test: **shapiro.test(x)**

H0: Distribution IS Normal.

p > 0.05 → **Normal (Parametric OK)**

p < 0.05 → **Not Normal (Use Non-Parametric)**

B. Homogeneity of Variance (Homoscedasticity)

Test: **var.test(x,y)** (2 grps) or **bartlett.test(y~grp)**

H0: Variances are EQUAL.

p > 0.05 → **Equal Var (Student T OK)**

p < 0.05 → **Diff Var (Use Welch T)**

C. Independence

Are groups Paired (Before/After) or Independent?

4. Distributions & Outliers

Normal: Continuous, Bell shape. **rnorm()**

Poisson: Count data, Mean=Var. **rpois()**

Binomial: Success/Fail (n trials). **rbinom()**

Uniform: Equal prob. **runif()**

Outliers (Grubbs Test)

Pkg: `outliers`. **grubbs.test(x)**

H1: Highest/Lowest value is an outlier.

Theory "Traps" (Don't fall for these!)

p-value definition: It is NOT the probability that H0 is true. It is the probability of the DATA given H0 is true.

CI 95%: It does NOT mean "95% chance the true mean is in here." It means "If we repeated the study 100 times, 95 of the intervals would contain the true mean."

Correlation != Causation: High r (0.9) does not mean X causes Y. It could be a third variable (Z).

Non-Parametric: Does NOT mean "no assumptions." It means "no assumption of normal distribution." (Still assumes independence/randomness).

KEY DEFINITIONS: BASICS

Inference: Drawing conclusions about a population based on a sample.

Central Limit Theorem: Means of samples are Normally distributed, even if data isn't (if N > 30).

Standard Error (SE): Accuracy of the Mean estimate. (SD / sqrt(N)).

p-value: Prob. of seeing data AT LEAST this extreme if Null Hypothesis is true.

Type I Error (α): Rejecting H0 when it is actually True (False Positive).

Type II Error (β): Failing to reject H0 when it is False (False Negative).

Power (1-β): Probability of correctly finding an effect that exists.

Degrees of Freedom: Number of values free to vary (usually N-1).

5. STATISTICAL TEST SELECTION GUIDE

Goal / Data Type	Condition / Groups	Parametric (Normal)	Non-Parametric (Non-Normal)	R Function
Compare 1 Mean/Med (vs Reference)	Reference Value	**One-sample T-test**	**Wilcoxon Signed Rank**	**t.test(x, mu=v)** **wilcox.test(x, mu=v)**
	Proportion vs Ref	**Binomial / Prop Test**		**prop.test()** / **binom.test()**
Compare 2 Groups (Continuous Y)	Independent	**Student's T-test** (Welch if var unequal)	**Mann-Whitney U** (Wilcoxon Rank Sum)	**t.test(y~grp)** **wilcox.test(y~grp)**
	Paired (Before/After)	**Paired T-test**	**Wilcoxon Signed Rank**	**t.test(..., paired=TRUE)**
Compare > 2 Groups (Continuous Y)	Independent	**One-Way ANOVA** (Post-hoc: Tukey)	**Kruskal-Wallis** (Post-hoc: Dunn/Wilcox)	**aov(y~grp)** **kruskal.test(y~grp)**
	Repeated Measures	**Repeated ANOVA**	**Friedman Test**	**friedman.test()**
Relationship (2 Continuous Vars)	Linearity	**Pearson Correlation**	**Spearman Correlation**	**cor.test(x, y, method=...)**
Relationship (2 Categorical Vars)	Contingency Table	**Chi-Square Test** (or Fisher's Exact if n<5)		**chisq.test(table)** **fisher.test(table)**

PG 2: STATISTICAL MODELING (COURS 3)

1. Linear Regression (OLS)

Model: \(Y = \beta_0 + \beta_1 X + \epsilon \)
Assumption: Residuals must be Normal & Homoscedastic.
Code: **model <- lm(Y ~ X1 + X2, data=df)**
Selection: **stepAIC(model)** (Pkg: MASS). Back/Forward.

Output Interpretation (summary(model))

Estimate: Slope. Unit change in Y for 1 unit X.

Pr(>|t|): Is variable significant? (< 0.05).

Multiple R²: % Variance explained by model.

Adj R²: Penalized for complexity. Use for model comparison.

F-statistic: Is the WHOLE model significant?

Diagnostics (plot(model))

Resid vs Fitted: Linearity check (Should be flat line).

Normal Q-Q: Normality of resid (Points on line).

Scale-Location: Homoscedasticity (Flat line).

Cook's Dist: Influential points/outliers.

Multicollinearity

If vars strongly correlated, estimates are unstable.
Check **VIF** (Pkg: `car`). If VIF > 5, remove variable.

2. The ANOVA Family

One-Way: 1 Factor (Cat) → Y (Cont). **aov(Y~Gr)**
Two-Way: 2 Factors. Checks Main Effects + **Interaction**.
Code: **aov(Y ~ A * B)** same as **aov(Y ~ A + B + A:B)**
Interaction (A:B): Does effect of A depend on B?

If Global P < 0.05 → Do **Post-hoc** (TukeyHSD).

Extensions

MANOVA: Multiple Y variables. **manova(cbind(Y1,Y2)~X)**.

ANCOVA: ANOVA + Continuous Covariate (to control bias).

Assumptions: No multicollinearity for MANOVA. Homogeneity of slopes for ANCOVA.

3. Non-Linear Models

Use when relationship isn't a straight line.

Quadratic: \(Y = X^2 + X \). **lm(Y ~ poly(X,2))**

Logarithmic: Rate decreases. **lm(Y ~ log(X))**

Exponential: Growth. **lm(Y ~ exp(X))**

Answer Phrasing Templates (Memorize!)

Rejecting H0 (Significant):

"Since p-value (0.003) < α (0.05), we reject H0. We conclude there is a statistically significant difference in [Variable Y] between [Group A] and [Group B]."

Failing to Reject H0 (Not Sig):

"Since p-value (0.12) > α (0.05), we fail to reject H0. We do not have sufficient evidence to demonstrate a difference..."

Interpreting OR/HR (e.g., 1.5):

"The [Risk/Odds] of [Event] is 1.5 times higher in the [Treatment Group] compared to [Control], holding other variables constant."

Interpreting Linear Model (\(\beta\) = -3.2):

"For every 1 unit increase in [X], [Y] decreases by 3.2 units on average."

R Code "Gotchas" & Fixes

Factor Error: If R treats numbers as categories or vice versa.
`df$grp <- as.factor(df$grp)` (Fixes "numeric" error in ANOVA).

Missing Data: Functions failing?
Add: `na.rm = TRUE` (mean/sd) or `use = "complete.obs"` (cor).

Formula Syntax:
`Y ~ X` (Y depends on X).
`data = df` (Always specify dataset).

Logic Operators:
== (Comparison), = (Assignment).
`!` (Not), & (And), | (Or).

Square Brackets:
`df[row, col]` → `df[1:5, ]` (First 5 rows).

4. Generalized Linear Models (GLM) & Evaluation

Used when Y is **NOT Normal** (Counts, Binary). Uses MLE (Max Likelihood).

A. Model Comparison (Goodness of Fit)

\(R^2\) is NOT used. Use these instead (Compare Model A vs B):

AIC (Akaike): Balances fit vs complexity.
Lower is better.

BIC (Bayesian): Penalizes variables more strictly.
Lower is better.

Deviance:
Null Deviance: Worst model (Intercept only).
Residual Deviance: Your model. (Should be much lower than Null).

B. Poisson (Count Data)

Y = 0, 1, 2... (e.g., # of events).
Code: **glm(Y ~ X, family = "poisson")**
CRITICAL CHECK: Overdispersion
If \(\text{Residual Dev} / \text{DF} \gg 1 \), assumption violated.
Fix: **family="quasipoisson"** or **glm.nb()** (Neg. Binomial).

C. Logistic (Binary Data)

Y = 0/1 (e.g., Sick/Healthy). Link=Logit.
Code: **glm(Y ~ X, family = "binomial")**
Coefficients: Log-Odds. Exponentiate for OR.
\(OR = \exp(\beta) \). (OR>1: Risk Increase).

D. Logistic Evaluation Metrics (Classification)

Confusion Matrix:

	Pred 1	Pred 0
Act 1	TP	FN
Act 0	FP	TN

Accuracy: \(\frac{TP+TN}{Total}\)

Sensitivity (TPR): \(\frac{TP}{TP+FN}\) (Detecting Sick)

Specificity (TNR): \(\frac{TN}{TN+FP}\) (Detecting Healthy)

ROC Curve & AUC:

- **ROC:** Plot of Sensitivity (Y) vs 1-Specificity (X).
- **AUC (Area Under Curve):** Measure of discrimination.
  0.5 = Random Guess (Diagonal Line).
  0.7 - 0.8 = Acceptable.
  > 0.8 = Excellent.
  1.0 = Perfect Separation.

PG 3: MULTIVARIATE ANALYSIS & ML (COURS 4)

1. Dimensionality Reduction (PCA)

Goal: Reduce variables (continuous) into Principal Components (PCs) while keeping variance.
Pkg: `FactoMineR` & `factoextra`.
Code: **PCA(df, scale.unit=TRUE, graph=T)** **ALWAYS SCALE!**

Scree Plot (Eigenvalues)
Bar chart of % variance explained by each PC.
Decision: Keep PCs until "Elbow" or >80% total variance.

Variables Plot (Circle)
Arrows same dir: Pos Correlation.
Opposite dir: Neg Correlation.
90° Angle: Uncorrelated.
Length: Quality of representation on axis.

Other Methods
MCA: For Categorical vars (Survey).
FAMD: Mixed (Cat + Cont).
FCA (CA): 2 Cat variables (Contingency).

Visual Analysis Cheat Keys

Boxplot
Thick Line: Median.
Box: IQR (25th-75th %).
Whiskers: 1.5 * IQR.
Dots: Outliers.

ROC Curve
X-axis: 1 - Specificity (FPR).
Y-axis: Sensitivity (TPR).
Diagonal: Random guess (AUC=0.5).
Top-Left: Perfect model.

Dendrogram (Cluster)
Height: Dissimilarity.
Cut: Draw horizontal line.
Count: # of vertical lines crossed = k clusters.

Interaction Plot
Parallel Lines: No Interaction.
Crossing Lines: Strong Interaction. (Effect of A flips based on B).

2. Unsupervised Learning: Clustering

Goal: Group data without labels.
K-Means
Partition into K groups. Centroid based.
Code: **kmeans(df, centers=K)**
Finding K (Elbow Method):
Plot Total Within Sum of Squares vs K. Look for bend.
fviz_nbclust(df, kmeans, method="wss")

3. Supervised Learning: Classification

Goal: Predict Class (Y) using inputs.
A. Linear Discriminant Analysis (LDA)
Finds linear combo to separate groups. Pkg: `MASS`.
Assumes normality & equal covariance.
lda(Y ~ ., data=df)
B. Decision Trees (CART)
Flowchart style. Pkg: `rpart`.
Pros: Visual, no normalization needed.
Cons: Overfitting.
Fix: Pruning (Cut branches). Check `plotcp()` for optimal CP value.

4. Model Evaluation Metrics

Based on **Confusion Matrix** (Predicted vs Actual).

	Pred 1	Pred 0
Act 1	TP	FN
Act 0	FP	TN

Sensitivity (Recall): $ TP / (TP+FN) $ - Ability to find Positives.
Specificity: $ TN / (TN+FP) $ - Ability to find Negatives.
Accuracy: $ (TP+TN) / Total $
PPV (Precision): $ TP / (TP+FP) $

PG 4: CLINICAL & ADVANCED ANALYSIS (COURS 5, 6)

1. Clinical Trial Design

Evidence Pyramid (Low to High)

Opinion → Case Series → Cross-Sectional → Case-Control → Cohort → **RCT** → **Meta-Analysis**.

Study Types

Observational: No intervention.
Cohort: Forward (Exp → Outcome). Risk factors.
Case-Control: Backward (Outcome → Exp). Rare diseases.
Cross-Sectional: Snapshot. Prevalence.

Experimental (RCT): Randomized. Gold Standard.
Parallel: Group A vs B.
Crossover: A then B (Washout period). Own control.

Bias & Terminology

Selection Bias: Groups differ at start (Fix: Randomize).

Performance Bias: Care differs (Fix: Blind/Double Blind).

Attrition Bias: Dropouts.

Phases: I (Safety), II (Dose/Efficacy), III (Large Comp), IV (Post-market).

Scientific Article: Bias Checklist

Scan the "Methods" and "Discussion" for these flaws:

- **Selection Bias:** Was the sample random? Or were they volunteers (healthier)?
  Check: Table 1 (Baseline Characteristics) differences.
- **Attrition Bias:** Did many people quit?
  Check: Flowchart. If >20% dropout, validity is compromised.
- **Analysis Method:**
  ITT (Intention to Treat): Analyzed in original groups (Gold Standard).
  PP (Per Protocol): Analyzed only if they finished meds (Biased).
- **Confounding:** Did they adjust for Age/Sex?
  Look for: "Adjusted Odds Ratio" or "Multivariate Analysis".
- **Conflict of Interest:** Who funded the study? (Pharma company?).

KEY DEFINITIONS: CLINICAL

Randomization: Allocating subjects by chance to reduce Selection Bias.

Double-Blind: Neither patient nor doctor knows the treatment (Reduces Performance Bias).

Censoring: In Survival Analysis, when an event (death) is NOT observed (drop out/end of study).

Hazard Ratio (HR): Instantaneous risk of event in Group A vs B.

Intention-To-Treat (ITT): Analyzing EVERYONE randomized, even if they quit (Prevents bias).

Confounding Variable: A 3rd var that correlates with both Exp and Outcome (e.g., Age).

Heterogeneity (I^2): In Meta-Analysis, how much results differ between studies due to real diffs vs chance.

Forest Plot: Graphical display of estimated results from a number of scientific studies.

Effect Size (Magnitude) & Power

Definition: The true magnitude of the difference. Unlike p-value, Effect Size is **independent of Sample Size**.

Cohen's Benchmarks (Memorize!)
Used in `pwr` package functions (e.g., `cohen.ES`)

Test	Metric	Small	Medium	Large
T-Test (Means)	**Cohen's d**	0.2	0.5	0.8
Correlation	**r**	0.1	0.3	0.5
ANOVA	**Cohen's f**	0.1	0.25	0.4
Chi-Square	**Cohen's w**	0.1	0.3	0.5

Calculation Logic

Formula (d): \(d = \frac{\text{Mean}_1 - \text{Mean}_2}{SD_{pooled}} \)
Package: `pwr`

```
pwr.t.test(d=0.5, power=0.8, sig.level=0.05)
```

The Trade-off:

- **Big Effect?** Need Small N.
- **Small Effect?** Need Huge N.
- **High Power (0.9)?** Need Big N.

2. Meta-Analysis

Meta-Analysis (Pkg: `meta`)

Synthesize multiple studies. Key Viz: **Forest Plot**.
Heterogeneity (I^2): Inconsistency across studies.
I^2 < 25% (Low) → Fixed Effects Model. I^2 > 50% (High) → Random Effects Model.
Bias Check: Funnel Plot (Asymmetry = Publication Bias).

Longitudinal Data (Mixed Models)

Goal: Analyze data with repeated measures on the same subjects over time (Correlated data).
Pkg: `nlme` Function: **lme()**

Fixed vs. Random Effects

- **Fixed Effect:** Population average (e.g., Treatment effect). Same for everyone.
- **Random Effect:** Individual deviation. Accounts for correlation within a subject.

R Syntax (Random Effects)

```
model <- lme(fixed = Y ~ Time, random = ..., data=df)
```

1. Random Intercept: `random = ~ 1 | ID`
Subjects start at different baselines, but change at same rate.

2. Random Slope + Int: `random = ~ Time | ID`
Subjects start differently AND change at different rates.

Covariance Structures
How are timepoints correlated?

- **Compound Symmetry:** Equal correlation between all time points.
- **AR(1):** Correlation decays over time (t1 is close to t2, far from t5).

3. Survival Analysis

Goal: Time until event (Death/Relapse). Handles **Censoring** (Event didn't happen yet).
Pkg: `survival`, `survminer`.

Kaplan-Meier (KM)

Non-parametric estimation of survival function S(t).
Curve: Steps down when event occurs.
Comparison: Log-Rank Test.
Code: **survdiff(Surv(time, status) ~ group)**
H0: Survival curves are identical.

Cox Proportional Hazards

Multivariate regression for survival.
Metric: Hazard Ratio (HR).
Code: **coxph(Surv(time, status) ~ Age + Sex)**
HR = 1: No effect.
HR > 1: Increases risk (Hazard). Bad.
HR < 1: Decreases risk (Protective). Good.
Assumption: Prop. Hazards (Check **cox.zph**).

4. Exam Tips: Analyzing the Article

1. **Identify Design:** Look for "Randomized" (RCT), "Retrospective" (Case-Control/Cohort).
2. **Identify PICO:** Population, Intervention, Control, Outcome (Primary Endpoint).
3. **Check Table 1:** Are groups balanced at baseline? (p-values should be > 0.05).
4. **Interpret Results:** Look at 95% Confidence Intervals.
   For Difference (Mean): If CI crosses 0 → Not Sig.
   For Ratio (OR/HR/RR): If CI crosses 1 → Not Sig.
5. **Critique:** Small N? Short duration? High dropout? Conflict of interest?